Kanishka Harwani

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EDUCATION

New York University, Brooklyn, NY, USA — 09/2023 – 05/2025
Master of Science, Mechatronics and Robotics — GPA: 3.667/4.0

Birla Institute of Technology and Science, Pilani, Hyderabad, Telangana, India — 08/2017 – 06/2021
Bachelor of Engineering, Electrical and Electronics Engineering — GPA: 6.75/10.0

TECHNICAL SKILLS

- **Programming Languages:** Python, Embedded C, C++
- **Robotics & Embedded Systems:** Autonomous Navigation, Sensor Fusion, Embedded Control Systems, IoT Systems
- **Frameworks & Libraries:** ROS 2, OpenCV, TensorFlow, PyTorch, Scikit-learn, Stable-Baselines3, Gym
- **Hardware & Electronics:** PCB Design & Bring-up (sensor and control boards), LTspice Circuit Simulation, Embedded Firmware Development (ESP32, real-time sensor interfacing)
- **Design & Simulation Tools:** MATLAB, Simulink, Fusion 360, Eagle CAD, KiCAD, LTspice
- **Development & OS Tools:** Linux (Ubuntu), Git, Jupyter Notebook, VS Code, CLion, PyCharm, Arduino IDE
- **Communication Protocols:** UART, SPI, I^2C, CANBUS, Bluetooth, Wi-Fi

WORK EXPERIENCE

Research Scholar Intern, New York University, MAE Department, Brooklyn, NY — 09/2025 - Present
- Designed and prototyped a one-wheel-based reaction-wheel-driven robotic platform, leading mechanical design, electronics integration, and embedded software development for a real-time balance control system.
- Implemented embedded real-time control and sensor fusion on ESP32 using IMU, TOF sensors, encoders, and CAN-based motor control via ODrive, enabling telemetry, control commands, and data acquisition over Wi-Fi using TCP/UDP communication.

Systems Engineer Intern, Gen Auto AI, Piscataway, NJ — 11/2025 - 02/2026
- Designed and integrated mechanical and electrical systems for an autonomous vehicle test-bed, including sensors, cameras, LiDAR, compute units, and power systems, supporting rapid prototyping and validation of autonomy algorithms.
- Developed custom mounting hardware and a drive control board to control steering, throttle, and braking via CAN bus, enabling system-level hardware integration including compute units, power distribution, and sensor mounting architecture.

Robotics Specialist, Young Tinker Academy, Bhubaneswar, India — 01/2022 - 08/2022
- Designed and developed educational robots using Fusion 360, EagleCAD, and TinkerCAD, integrating 3D design, electronics, and programming for student learning.
- Led a software team to build Bluetooth control apps and a Blockly-based coding platform, optimizing robot and packaging design to reduce material waste by 18% and production costs by 34

PROJECTS

Reaction Wheel Based Balance Bot: — 09/2025 - Present
- Developed a real-time balance control stack for a single-wheel electric skateboard platform, enabling pitch stabilization through movable mass actuation and a BLDC-driven reaction system controlled via ODrive Micro over CAN.
- Built embedded state estimation and control stack on ESP32 using IMU and 5 ToF sensors, applying ZUPT, quaternion-based orientation, and Kalman filtering for improved pose estimation and stability.

Hardware and Embedded Systems for Semi-Autonomous Electric Go-Kart: — 11/2025 - Present
- Led hardware and embedded systems integration for a semi-autonomous electric go-kart, implementing steering, throttle, and braking actuation using motor drivers and CAN-based control.
- Developed and deployed embedded firmware for CAN communication and system interfacing, rewiring the dashboard and powertrain ESC to integrate LiDAR, cameras, and a modular sensor stack for real-time data acquisition.

ROS2-Enabled Micro Mobility Robot Platform: — 01/2026 - Present
- Designed and built a compact ROS 2–enabled mobile robot platform from scratch using ESP32 with micro-ROS, supporting teleoperation over Bluetooth/Wi-Fi and acting as a ROS node for algorithm development and testing.
- Integrated IMU, motor drivers, and onboard feedback systems (addressable LEDs, buzzer), enabling real-time sensing and control; currently developing a custom PCB to expand sensor integration and improve system modularity.

AMR Test Platform: — 09/2024 - 05/2025
- Built a four-wheel-drive autonomous mobile robot equipped with mmWave radar, cameras, IMU, and GPS for safety testing in human-robot shared environments.
- Spearheaded the development of a ROS2-based framework for real-time sensor fusion and safety algorithms, enabling comparative testing of radar vs. camera perception and autonomous vs. remote-controlled operation.

PUBLICATION

"Design and Analysis of Dynamics, Control and Simulation of Y-4 Quadrotor Structure for Hybrid Aerial Vehicle," IEEE INCET 2020.